

Name: File  
Class: \_\_\_\_\_

## SYDNEY TECHNICAL HIGH SCHOOL



## MATHEMATICS

### HSC ASSESSMENT TASK 3

**JUNE 2005**

*Time Allowed: 70 minutes*

#### Instructions

- Write your name and class at the top of this page, and at the top of each answer sheet.
- At the end of the examination, this examination paper must be attached to the front of your answers.
- Attempt all questions.
- All necessary working must be shown. Marks will be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.

| Q1 | Q2 | Q3 | Q4 | Q5 | Total |
|----|----|----|----|----|-------|
| 12 | 12 | 10 | 11 | 11 | 56    |

**Question 1 (12 marks)**

a) Express  $315^\circ$  in radians in terms of  $\pi$  (1)

b) Find the exact value of  $\cos \frac{5\pi}{4}$  (1)

c) Differentiate

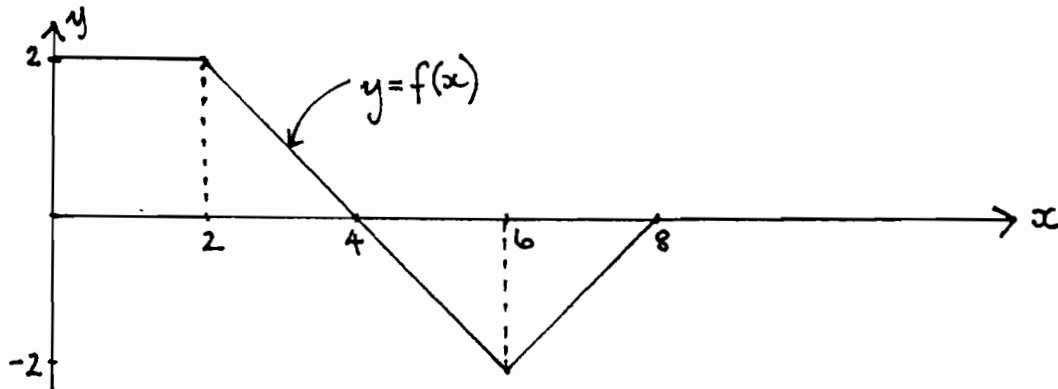
i)  $y = 2\sqrt{x}$  (1)

ii)  $y = \frac{1}{x-1}$  (1)

iii)  $y = \frac{x+1}{x-1}$  (2)

d) Find the primitive of  $(2x+1)^4$  (2)

e) The function  $y = f(x)$  is sketched below



i) Evaluate  $\int_0^8 f(x) dx$  (1)

ii) Find the area enclosed by  $y = f(x)$  the  $x$  axis,  $x = 0$  and  $x = 8$  (1)

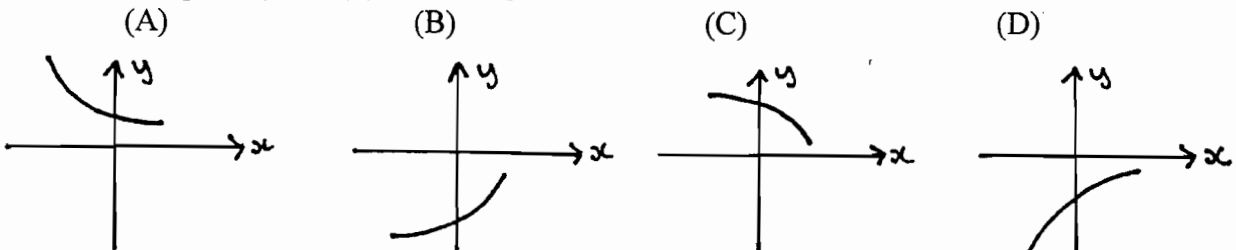
For parts f) and g) write the correct letter on your answer sheets.

f) Which one of the following is equal to  $\int k dx$ , where  $k$  is a constant.

(A)  $kx + C$  (B)  $\frac{k^2}{2} + C$  (C)  $\frac{k^2 x}{2} + C$  (D)  $\frac{kx^2}{2} + C$  (1)

g) For a given function  $y = f(x)$ ,  $f(0) < 0$ ,  $f'(x)(0) > 0$  and  $f''(x)(0) > 0$ .

The graph of  $y = f(x)$  in this region would look like: (1)



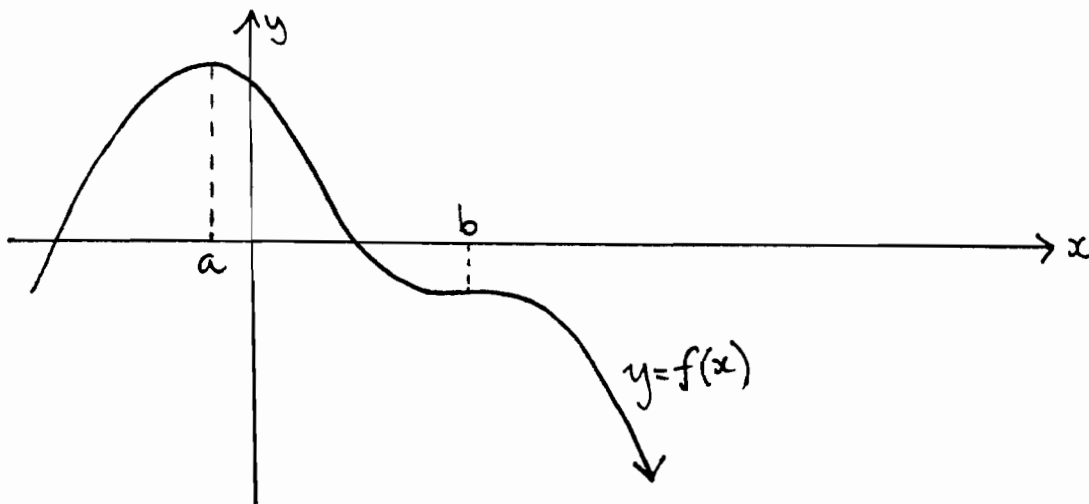
**Question 2 (12 marks) (Start a new page)**

a) Sketch  $y = 2 \sin \frac{x}{2}$  for  $-2\pi \leq x \leq 2\pi$  (2)

b) Solve  $\sin x = -\frac{\sqrt{3}}{2}$  for  $0 \leq x \leq 2\pi$  (2)

(Solution(s) must be in radians)

c) Given the sketch of  $y = f(x)$  (2)



Make a neat sketch of  $y = f'(x)$ . Indicate a and b on your sketch.

d) Consider the curve  $y = x^3(4 - x)$  in the domain  $-2 \leq x \leq 4$

i) Find the stationary points and determine their nature (3)

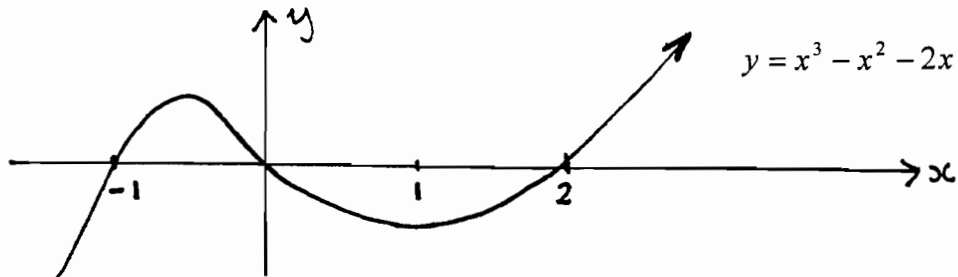
ii) Find any points of inflexion (1)

iii) Sketch the curve in the domain  $-2 \leq x \leq 4$ .

Show the co-ordinates of the end points (2)

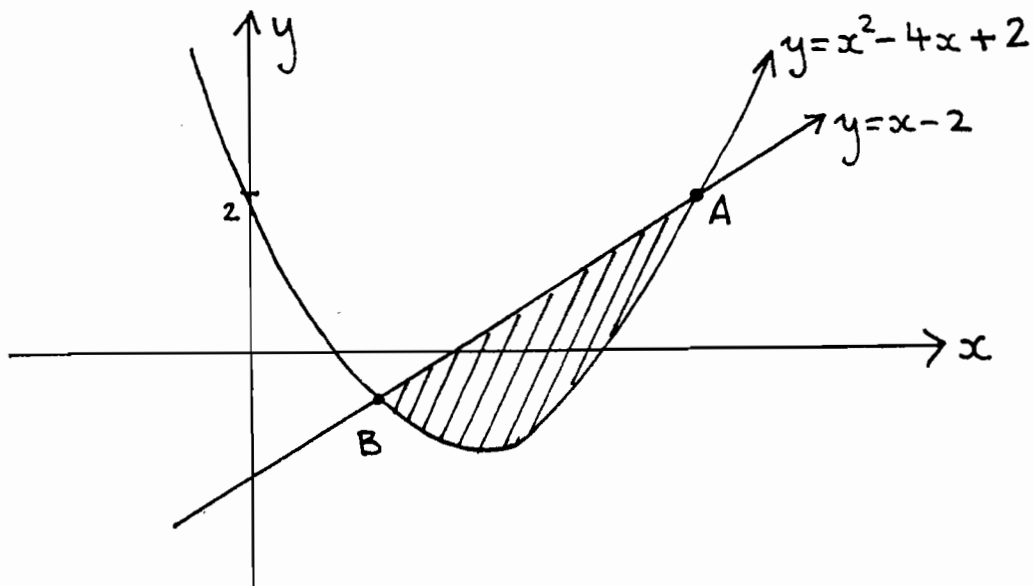
**Question 3 (10 marks) (Start a new page)**

- a) The curve  $y = x^3 - x^2 - 2x$  is sketched below (4)



Find the area enclosed by the curve and x axis.

- b) The curves  $y = x^2 - 4x + 2$  and  $y = x - 2$  are sketched below.



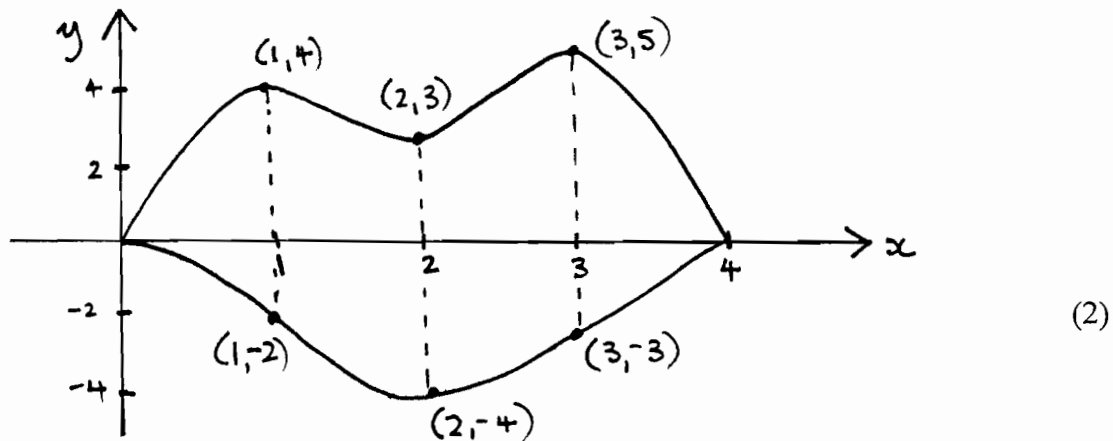
- i) Find the x co ordinate for A and B. (2)  
ii) Find the area of the enclosed shaded region. (4)

**Question 4** (11 marks) (Start a new page)

- a) The area below the graph  $y = \frac{1}{x}$  in the first quadrant between  $x = 1$  and  $x = 4$  is rotated around the  $x$  axis. What is the volume of the generated solid? (3)  
(answer in terms of  $\pi$ ).

- b) For a certain curve  $\frac{d^2y}{dx^2} = 6x - 10$ . Find the equation of the curve if it passes through the point  $(1, 1)$  with gradient  $-1$ . (3)

c)

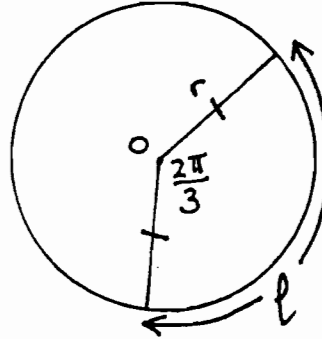


Use the trapezoidal rule and 5 function values to find the area of the enclosed region.

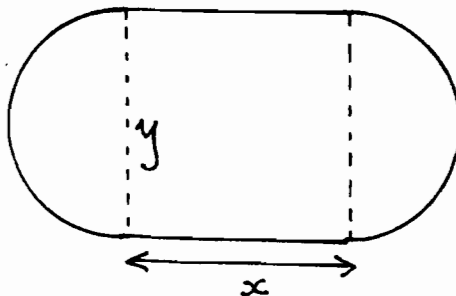
- d) If  $\int_1^b (2x-1)dx = 6$  and  $b > 0$ . Find  $b$ . (3)

**Question 5 (11 marks) (Start a new page)**

- a) A circular pond has an area of  $615.75m^2$ . It is divided into sectors one of which contains a central angle of  $\frac{2\pi}{3}$  radians.



- Find the radius  $r$  of the circle (to 1 dec pl ) (1)
  - Use  $r$  to find the length  $l$  of the sector ( to 1 dec pl ) (1)
  - Find the area of the sector ( to 1 dec pl ) (1)
- b) Solve  $\sin x + \cos x = 0$  for  $-\pi \leq x \leq \pi$  (solution must be in radians). (2)
- c) A running track of length 400 metres is designed using two sides of a rectangle and two semicircles, as shown. The rectangle has length  $x$  metres and the semicircles each have diameter  $y$  metres.



- Show that  $x = \frac{1}{2}(400 - \pi y)$ . (1)
- The region inside the track will be used for field events.  
Show that its area is  $A = 200y - \frac{\pi y^2}{4}$  (2)
- Hence find the maximum area that may be enclosed to the nearest whole

**Question 1**

a)  $315^\circ = 315 \times \frac{\pi}{180} = \underline{\underline{\frac{7\pi}{4}}}$

b)  $\cos \frac{5\pi}{4} = \cos(\pi + \frac{\pi}{4})$   
 $= -\cos \frac{\pi}{4}$   
 $= \underline{\underline{-\frac{1}{\sqrt{2}}}}$

c) i)  $\frac{d}{dx}(2\sqrt{x}) = 2 \times \frac{1}{2} x^{-1/2}$   
 $= \underline{\underline{\frac{1}{\sqrt{x}}}}$

ii)  $\frac{d}{dx}(x-1)^{-1} = -(x-1)^{-2}$   
 $= \underline{\underline{\frac{-1}{(x-1)^2}}}$

iii)  $u = x+1 \quad v = x-1$   
 $u' = 1 \quad v' = 1$   
 $\frac{dy}{dx} = \frac{(x-1) - (x+1)}{(x-1)^2}$   
 $= \underline{\underline{\frac{-2}{(x-1)^2}}}$

d)  $\int (2x+1)^4 dx = \underline{\underline{\frac{(2x+1)^5}{10} + c}}$

e) i)  $\int_0^8 f(x) dx = 4 - \frac{2 \times 2}{2}$   
 $= \underline{\underline{2}}$

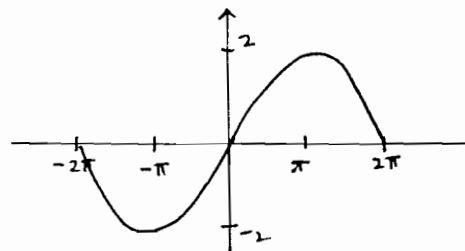
ii) area =  $4 + 3(2)$   
 $= \underline{\underline{10 \text{ unit}^2}}$

f) A

g) B

**Question 2**

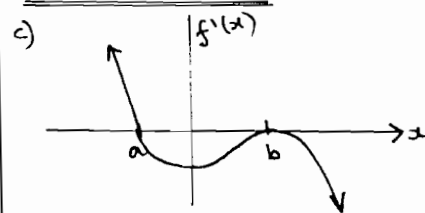
a) amplitude =  $\underline{\underline{2}}$   
 period =  $\frac{2\pi}{1/2} = \underline{\underline{4\pi}}$



b)  $\sin x = -\frac{\sqrt{3}}{2}$   
 acute  $x = \frac{\pi}{3}$   
 $\therefore x = \frac{4\pi}{3}, \frac{5\pi}{3}$

|   |   |
|---|---|
| S | A |
| T | C |

✓ | ✓



d)  $y = x^3(4-x) \quad -2 \leq x \leq 4$   
 end pts  $(-2, -48)$   $(4, 0)$

i)  $y = 4x^3 - x^4$   
 $\frac{dy}{dx} = 12x^2 - 4x^3$   
 $\frac{d^2y}{dx^2} = 24x - 12x^2$

st pt  $12x^2 - 4x^3 = 0$   
 $4x^2(3-x) = 0 \quad x = 0, 3$

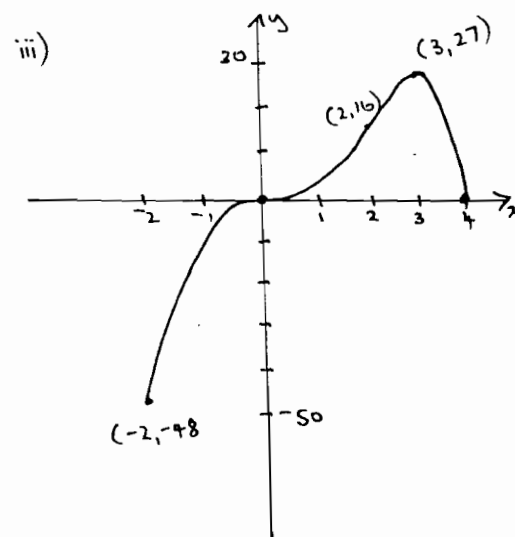
$\therefore (0, 0) \quad y'' = 0$   
 $(3, 27) \quad y'' < 0 \text{ max}$

test concavity at  $(0, 0)$

|           |       |     |     |
|-----------|-------|-----|-----|
| concavity | $x$   | -1  | 1   |
| change    | $y''$ | -ve | +ve |

$\therefore$  horizontal point of inflection

ii)  $\frac{d^2y}{dx^2} = 0 \quad 24x - 12x^2 = 0$   
 $12x(2-x) = 0$   
 $x = 0 \quad (0, 0) \text{ horz pt inf}$   
 $x = 2 \quad (2, 16) \text{ pt inf}$   
on continuous curve



**Question 3**

a)  $A = \int_{-1}^0 x^3 - x^2 - 2x dx + \left| \int_0^2 x^3 - x^2 - 2x dx \right|$   
 $= \left[ \frac{x^4}{4} - \frac{x^3}{3} - x^2 \right]_{-1}^0 + \left| \left[ \frac{x^4}{4} - \frac{x^3}{3} - x^2 \right]_0^2 \right|$   
 $= \left[ 0 - \left( \frac{1}{4} - \frac{1}{3} - 1 \right) \right] + \left| \left( \frac{16}{4} - \frac{8}{3} - 4 \right) - 0 \right|$   
 $= \frac{5}{12} + 2\frac{2}{3}$   
 $= \underline{\underline{3\frac{1}{12}}}$

b) i)  $x^2 - 4x + 2 = x - 2$   
 $x^2 - 5x + 4 = 0$   
 $(x-4)(x-1) = 0$   
 $x = 4, 1$   
A(4, 2) B(1, -1)

ii)  $A = \int_1^4 (x-2) - (x^2 - 4x)$   
 $= \int_1^4 (x-2-x^2+4x)$   
 $= \int_1^4 (-x^2+5x-4)$   
 $= \left[ -\frac{x^3}{3} + \frac{5x^2}{2} - 4x \right]_1^4$   
 $= \left[ \left( -\frac{64}{3} + 40 - 16 \right) - \left( -\frac{1}{3} + \frac{5}{2} - 4 \right) \right]$   
 $= 2\frac{2}{3} - \left( -1\frac{5}{6} \right)$   
 $= \underline{\underline{4\frac{1}{2} \text{ unit}^2}}$

**Question 4**

a)   $V = \pi \int_1^4 \left( \frac{1}{x} \right)^2 dx$   
 $= \pi \int_1^4 \frac{1}{x^2} dx$   
 $= \pi \left[ -\frac{1}{x} \right]_1^4$   
 $= \pi \left[ -\frac{1}{4} - (-1) \right]$   
 $= \underline{\underline{\frac{3\pi}{4} \text{ unit}^3}}$

$$b) \frac{d^2y}{dx^2} = 6x - 10$$

$$\frac{dy}{dx} = 3x^2 - 10x + c$$

$$\text{at } x=1 \quad \frac{dy}{dx} = -1$$

$$-1 = 3 - 10 + c$$

$$-1 = -7 + c$$

$$\therefore c = 6$$

$$\frac{dy}{dx} = 3x^2 - 10x + 6$$

$$y = x^3 - 5x^2 + 6x + k$$

$$\text{sub } (1, 1)$$

$$1 = 1 - 5 + 6 + k$$

$$1 = 2 + k$$

$$k = -1$$

$$\therefore y = x^3 - 5x^2 + 6x - 1$$

$$c) A_7 = \frac{1}{2} [0 + 0 + 2(6 + 7 + 8)]$$

$$= 21 \text{ unit}^2$$

use table  
to get  
above

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| x | 0 | 1 | 2 | 3 | 4 |
| y | 0 | 6 | 7 | 8 | 0 |

$$d) \int_1^b (2x-1) dx = \left[ x^2 - x \right]_1^b$$

$$= (b^2 - b) - (1 - 1)$$

$$\therefore b^2 - b = 6$$

$$b^2 - b - 6 = 0$$

$$(x-3)(x+2) = 0$$

$$b = 3 \quad b = -2 \quad \text{but } b > 0$$

$$\therefore b = 3 \text{ only}$$

### Question 5

$$a) i) A = \pi r^2 \quad \therefore 615.75 = \pi r^2$$

$$r = 14.0 \text{ cm}$$

$$(1 \text{ dec pl})$$

$$ii) l = r \theta$$

$$= 14 \times 2\pi/3$$

$$= 29.3 \text{ cm}$$

$$iii) A = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} \times 14^2 \times \frac{2\pi}{3}$$

$$= 205.3 \text{ cm}^2$$

$$b) \sin x = -\cos x$$

$$\tan x = -1$$

$$\text{acute } x = \pi/4$$

$$x = \left( \frac{3\pi}{4} \right) - \frac{\pi}{4}$$

|     |     |
|-----|-----|
| ✓ S | A   |
| T   | C ✓ |

$$c) i) 400 = 2\pi \frac{y}{2} + 2x$$

$$\frac{400 - \pi y}{2} = x$$

$$\therefore x = \frac{1}{2} (400 - \pi y)$$

$$ii) A = \pi x \left( \frac{y}{2} \right)^2 + xy$$

$$= \frac{\pi y^2}{4} + y \frac{1}{2} (400 - \pi y)$$

$$= \frac{\pi y^2}{4} + 200y - \frac{\pi y^2}{2}$$

$$= 200y - \frac{\pi y^2}{4}$$

$$iii) \frac{dA}{dy} = 200 - \frac{2\pi y}{4}$$

$$= 200 - \frac{\pi y}{2}$$

$$\frac{d^2A}{dy^2} = -\frac{\pi}{2} < 0 \quad \therefore \text{max}$$

$$\text{at pt } \frac{dA}{dy} = 0$$

$$200 = \frac{\pi y}{2}$$

$$y = \frac{400}{\pi}$$

$$\therefore \text{MAX area} = 200 \times \frac{400}{\pi} - \frac{\pi}{4} \left( \frac{400}{\pi} \right)^2$$

$$= \frac{80000}{\pi} - \frac{40000}{\pi}$$

$$= \frac{40000}{\pi}$$

$$\text{Max Area} = \frac{12732 \text{ m}^2}{\pi}$$